MADHAV JOSHI

+91 9823289767 ⋄ madhavjoshi2819@gmail.com

EDUCATION

Bachelor of Engineering in Mechanical Engineering June 2023 - May 2027
Dr. D Y Patil Institute of Engineering, Management And Research
Savitribai Phule Pune University, CGPA: 8.1/10.00

HSC, Class XII June 2022 - April 2023
Abhishek Jr College of Arts, Commerce and Science, 74.87%

SSC, Class X June 2020 - March 2021
Somalwar Highschool, Nikalas Branch, Nagpur, 83.47%

SKILLS AND INTERESTS

Interests	Product Development/Management, Design, Automobile, CAD/CAE, Optimization, Fluid Mechanics, Robotics, Modeling and Simulation
Design & Simulation Software	Basic AUTOCAD, SOLIDWORKS, ANSYS (Static Structural, Static Thermal), MSC AdamsDriveline, MATLAB COMSOL Multiphysics, KISSsoft, CATIA Magic

PROJECTS

Product Development: Custom CV Joint March 2025 - September 2025
SAEINDIA eBAJA, Genesis 16 Motorsports

- Performed OEM reverse-engineering and material evaluation to define design requirements.
- Designed and developed a custom CV joint tailored to the application.
- Integrated the CV joint directly with the gearbox output shaft, eliminating the need for an external CV joint assembly and reducing system complexity.
- Increased articulation angle from **27° to 37°**, enabling an additional **1 inch of wheel travel** in full droop conditions.

High-Performance Motor Tuning & Drivetrain Optimization August 2025 - December 2025
SAEINDIA eBAJA, Genesis 16 Motorsports

- Achieved **10–12% increase in powertrain performance** through advanced **motor (MCU) tuning**, optimizing FOC parameters, torque mapping, and ramp up/down tuning.
- **Eradicated ~100% belt slip losses** through precise motor tuning and drivetrain optimization, significantly improving torque delivery and acceleration.
- Validated improvements using **MATLAB-based simulations and models** and real-world testing, ensuring data-driven optimization of drivetrain efficiency and response.

Advanced Vehicle Dynamics Measurement & DAQ Systems Development August 2024
SAEINDIA eBAJA, Genesis 16 Motorsports

- Developed a **laser-based acceleration measurement system** using Raspberry Pi, NodeMCU, and Arduino; processed data via Python/MATLAB with noise filtering and vibration isolation to generate accurate performance metrics.
- Built a **vehicle attitude DAQ system** using ESP32 and MPU6050 IMU to measure real-time roll, pitch, and yaw; implemented sensor fusion and filtering for improved accuracy under off-road conditions.
- Analyzed **real-time data** across systems to evaluate drivetrain response, vehicle stability, handling characteristics, and overall off-road performance.
- Correlated **simulated models with real-world test data** to validate results and improve understanding of vehicle dynamics, suspension behavior, and drivetrain response.

POSITION OF RESPONSIBILITY

Powertrain and HV Lead, eBAJA March 2025 - Present
SAEINDIA eBAJA, Genesis 16 Motorsports

- Designed, optimized and developed **complete drivetrain system**, including custom CV joint (up to **37° articulation**), belt transmission, 2-stage gearbox, and differential integration for eBAJA 2026.
- Led **motor–transmission integration and advanced MCU tuning** (FOC parameters, throttle mapping, torque limits, regen, ramp control), along with CAN-based calibration, BMS tuning, and HV system integration.
- Optimized **pulley ratios and drivetrain dynamics** to improve traction, minimize losses, and enhance real-world power delivery based on test data.
- Achieved **AIR 3 in Acceleration** and **AIR 4 in Sled Pull**, driven by **powertrain performance optimization and tuning**.

Manufacturing Engineering Lead June 2024 - Present
SAEINDIA eBAJA, Genesis 16 Motorsports

- Driving **manufacturing-centric design**, ensuring production efficiency, structural robustness, and seamless subsystems–chassis integration.
- Gaining hands-on industrial exposure (MIDC) in machining, fabrication workflows, tolerance control, and assembly practices; translate insights into improved manufacturing quality.
- Leading **shop-floor execution and fabrication planning**, managing task delegation, precision-critical processes, and ensuring timely, standards-compliant build completion.

EXTRA-CIRRUCULAR

- Participated in SAE INDIA eBAJA, NATRAX, Indore,
  Secured **AIR-4** Overall Event, **AIR-2** Overall Dynamic Events,
  AIR-3 Acceleration, **AIR-4** Sledpull,
  AIR-4 Maneuverability, **AIR-5** Durability — January 2026, 25
- Participated in **Ansys 1-Day Workshop**, Secured 3rd Rank in quiz,
  DYP IU, Akurdi — April 2026
- Completed **Altair Hypermesh- FEA POST-PROCESSING**
  Associate Specialist Certification. — March 2025
- Secured Rank 1,college round-**SIH 2025**, presented study on modifications
  in sports-helmets using EPLA material through additive manufacturing — September-2025
- **Assistant Technical Evaluation Judge** at Morphine Motorsports
  Karting Competition, Pune — September-2025